

Labwork 1: Gradient Descent

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1 Algorithm implementation

In this lab, we implement gradient descent from scratch. Here is the update rule:

$$x_{new} = x_{old} - r \cdot f'(x_{old})$$

where:

- r is the learning rate
- $f'(x)$ is the derivative of the function

The algorithm starts from an initial value and iteratively updates x .

2 Effect of learning rate

The learning rate r plays a crucial role in the behavior of gradient descent:

- **Small learning rate** ($r = 0.01$): Convergence is slow and requires many iterations.
- **Moderate learning rate** ($r = 0.1$): Efficient convergence to the minimum.
- **Large learning rate** ($r = 1.0$): The algorithm may miss the minimum or even diverge.

Thus, choosing an appropriate learning rate is important for both speed and stability.